SENG 473 Information Security

Final-Project

(Deadline 24/05/2026 23:59)

There are five available topics for the final project, and you may choose any one of them. Within your selected topic, you are free to plan, design, develop, and present the project in your own way. You are also encouraged to explore and incorporate any relevant aspects, features, or approaches related to the chosen topic.

1. Zero Trust Network Implementation

- You will build a secure network where "internal" access is never assumed. You will implement identity-based controls to ensure only authenticated users can access specific services, regardless of their location.

2. Automated DevSecOps Pipeline

- You will create an automated "security gate" for software. Your pipeline must scan code for secrets and vulnerabilities automatically, blocking unsecure updates from being deployed.

3. Web App Penetration & Hardening

- You will act as security engineers. You'll first break into a vulnerable web application and then write the code necessary to patch those holes and secure the system.

4. Ransomware Response & Recovery

- You will simulate a ransomware attack by writing a controlled encryption script. Your goal is to design and demonstrate a recovery system that can restore data without paying the "attacker."

5. Blockchain Identity Management

- You will build a decentralized login system. Instead of a central database that can be leaked, you'll use blockchain technology to verify user identities through secure, cryptographic hashes.

Submission
The submission for this final project will be only an **unlisted YouTube video link** that you recorded clearly demonstrating two things:

- **Code Walkthrough:** A brief explanation of how you approached the topic on your code.
- **Live Demonstration:** Shows every detail that you applied onto your project

Team Rules
- You may work in teams of **2–3 members** (maximum 3).
- Each team member must submit the project link individually to the LMS system.

Note: You must mention your team's name in the recorded video.